Biodiversity and the Western Ghats

Knowledge document 035 of 100

Introduction to Biodiversity

Biodiversity, short for biological diversity, is the variety of all life on Earth, including plants, animals, fungi and microorganisms, the genes they contain and the ecosystems they form. It is usually discussed at three levels: genetic diversity (variation within a species), species diversity (the number of different species) and ecosystem diversity (the variety of habitats such as forests, wetlands and coral reefs). Scientists have described about 2 million species, but the total may be many millions more. Biodiversity provides food, medicine, clean air and water, pollination, soil fertility and climate regulation, and has cultural value. The Convention on Biological Diversity was adopted at the Rio Earth Summit in 1992, and the International Day for Biological Diversity is celebrated on 22 May. Biodiversity is being lost quickly because of habitat destruction, climate change, pollution, overexploitation and invasive species.

Biodiversity Hotspots

The concept of a biodiversity hotspot was introduced by the British ecologist Norman Myers in 1988. A region qualifies as a hotspot if it contains at least 1,500 species of endemic vascular plants, which means species found nowhere else, and has lost at least 70 percent of its original natural vegetation. There are about 36 hotspots worldwide, covering a small share of land but holding more than half of all endemic plant species. India has four hotspots: the Himalaya, the Indo-Burma region, the Western Ghats and Sri Lanka, and Sundaland (which includes the Nicobar Islands).

The Western Ghats

The Western Ghats, also called the Sahyadri, are a mountain range running about 1,600 kilometres along the western edge of the Indian peninsula, parallel to the Arabian Sea coast, from the Tapi river near the Gujarat-Maharashtra border to Kanyakumari at the southern tip. They pass through six states: Gujarat, Maharashtra, Goa, Karnataka, Kerala and Tamil Nadu, and cover about 160,000 square kilometres. The highest peak is Anamudi in Kerala, at about 2,695 metres. The range is older than the Himalaya, is the source of many peninsular rivers such as the Godavari, Krishna and Kaveri, and strongly influences the monsoon by forcing moist winds to rise and shed rain.

Why Are the Western Ghats a Biodiversity Hotspot?

The Western Ghats are a global biodiversity hotspot for several reasons. They receive very heavy monsoon rainfall and have a wide range of altitudes and climates, from tropical evergreen forests and moist deciduous forests to montane grasslands called shola, which together create many habitats. The mountains have been isolated for millions of years, which allowed many species to evolve nowhere else. As a result, the region has an extraordinary number of endemic species: over 5,000 species of flowering plants, with about a third endemic, about 139 mammal species, 508 bird species, and 179 amphibian species, of which most are endemic. More than 300 globally threatened species live here. The region is also home to the largest wild population of Asian elephants and a large number of tigers. New species are discovered here regularly, especially amphibians and insects.

Animals Found Only in the Western Ghats

Two animals found only in the Western Ghats are the Nilgiri tahr and the lion-tailed macaque. The Nilgiri tahr is a mountain goat living on high grassy cliffs of the Nilgiri and Anamalai hills, and it is the state animal of Tamil Nadu. The lion-tailed macaque is a black monkey with a silvery-white mane found in the rainforests of the southern Western Ghats. Other endemic species include the Malabar large-spotted civet, the Nilgiri marten, the Malabar grey hornbill, the Nilgiri wood pigeon, the purple frog, which spends most of its life underground, and many species of caecilians and shrub frogs. Tamil Nadu's protected areas in the Ghats include the Mudumalai, Anamalai and Kalakkad Mundanthurai Tiger Reserves and the Nilgiri Biosphere Reserve, India's first biosphere reserve, established in 1986.

When Were They Made a UNESCO Site?

The Western Ghats were inscribed as a UNESCO World Heritage Site in July 2012, at the 36th session of the World Heritage Committee in St. Petersburg, Russia. The site consists of 39 serial properties spread across the six states, including national parks, wildlife sanctuaries and reserve forests. In Tamil Nadu these include parts of the Kalakkad Mundanthurai and Mukurthi National Park areas. It is recognised as a natural site of outstanding universal value for its unique biological and ecological processes.

Threats and Conservation

The Western Ghats are threatened by deforestation, plantations, mining, dams, roads, tourism, climate change, landslides and conflict between humans and wildlife. In 2010 the Western Ghats Ecology Expert Panel headed by Madhav Gadgil recommended strong protection, and the High-Level Working Group chaired by K. Kasturirangan in 2013 proposed protecting about 37 percent of the region as ecologically sensitive. India protects biodiversity under the Wildlife Protection Act, 1972, the Biological Diversity Act, 2002, Project Tiger (1973) and Project Elephant (1992).

Challenges and Future

The global Kunming-Montreal Biodiversity Framework of December 2022 aims to protect 30 percent of land and sea by 2030. Community conservation, restoring habitats and sustainable development in the Western Ghats are important for the future.

Quick Facts

- A biodiversity hotspot has at least 1,500 endemic plant species and has lost 70 percent of its habitat.
- The Western Ghats run about 1,600 km along India's west coast through six states.
- Nilgiri tahr and lion-tailed macaque are found only in the Western Ghats.
- The Western Ghats became a UNESCO World Heritage Site in July 2012.
- India's four biodiversity hotspots are the Himalaya, Indo-Burma, Western Ghats and Sri Lanka, and Sundaland.